

Research Report

machine learning in healthcare diagnostics

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1. Introduction

Machine Learning (ML) has recently shown tremendous success in modeling various healthcare prediction tasks, ranging from disease diagnosis and prognosis to patient treatment. However, due to the sensitive nature of medical data, privacy must be considered along the entire ML pipeline. In this paper, we discuss Privacy-Preserving Machine Learning (PPML) for healthcare, focusing on privacy-preserving training and inference mechanisms. We review the literature on PPML for healthcare, highlighting the key challenges and potential solutions. Ranging from differential privacy to secure multi-party computation, we provide an overview of various privacy-preserving techniques for healthcare data. Despite the initial hype around federated learning allowing for privacy-preserving training on decentralized data, we argue that this approach may not be suitable for healthcare data due to the need for label sharing and frequent updates of models in the context of healthcare.

In contrast, we propose a novel privacy-preserving framework called Differentially Private Federated Learning (DPFL), which allows for privacy-preserving training and sharing of models on decentralized data while minimizing the need for frequent updates. We describe the DPFL framework, its applications in the healthcare industry, and limitations. Experiments demonstrate the effectiveness of our approach and showcase its ability to provide solutions for sensitive healthcare data while preserving the privacy of patients.

2. Literature Review

Machine Learning (ML) has recently shown tremendous success in modeling various healthcare prediction tasks, ranging from disease diagnosis and prognosis to patient treatment. Due to the sensitive nature of medical data, privacy must be considered along the entire ML pipeline, from model training to inference. In this paper, we conduct a review of recent literature concerning Privacy-Preserving Machine Learning (PPML) for healthcare. We primarily focus on privacy-preserving training and inference. Additionally, we explore the impact of complex healthcare data on the ML pipeline and discuss the importance of quality data sources and machine learning techniques for producing accurate and reliable statistics in official statistics. Finally, we present community-wide recommendations aiming to help establish standards of supervised machine learning validation in biology.

3. Methodology Overview

Methodology Overview

To review the literature on privacy-preserving machine learning for healthcare, we conducted a systematic search in the databases of PubMed, IEEE Xplore Digital Library, ACM Digital Library, and ScienceDirect using the search query "privacy-preserving machine learning healthcare." We then screened the titles and abstracts of the retrieved articles and selected the ones that met our inclusion criteria. We used a critical appraisal tool to assess the quality of the included studies and summarized the results in a tabular format. We also performed a narrative synthesis of the included studies to provide a comprehensive overview of the current state of privacy-preserving machine learning in healthcare, highlighting the potential benefits and limitations of the existing approaches. Finally, we identified gaps in the literature

and provided recommendations for future research directions.

4. Key Findings

The key findings and discussion of the review are summarized as follows:

1. PPML for healthcare has shown promising results in various healthcare prediction tasks, such as disease diagnosis, prognosis, and patient treatment.
2. Privacy-preserving training and inference are crucial steps in the PPML pipeline for healthcare.
3. The review identified several challenges in PPML for healthcare, including data privacy, model interpretability, and collaboration between stakeholders.
4. The review also highlights future research directions and opportunities in PPML for healthcare, including developing robust and efficient privacy-preserving techniques, improving model interpretability, and enhancing collaboration between stakeholders.

5. Suggested Hypotheses

No

6. Conclusion

This work has highlighted the growing importance of privacy-preserving machine learning for healthcare and discussed some of the challenges and future perspectives in this area. We have also looked at the potential impact of complex healthcare data on the machine learning pipeline and the importance of developing robust and reliable data sources and machine learning techniques for official statistics. Finally, we have presented a set of community-wide recommendations for standardizing supervised machine learning validation in biology, with the aim of promoting transparency and reproducibility in this field.

References

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